

练习五 力学基本定律（五）

1. $m\sqrt{gy_0}(1+\sqrt{2}), \frac{1}{2}mv_0$
2. lmv ;
3. (3);
4. (4);
5. (1) 子弹进入沙土后受力为 $-Kv$, 由牛顿定律:

$$-Kv = m \frac{dv}{dt}$$

$$-\frac{K}{m} dt = \frac{dv}{v}, \quad -\int_0^t \frac{K}{m} dt = \int_{v_0}^v \frac{dv}{v}$$

$$v = v_0 e^{-Kt/m}$$

$$\int_0^x dx = \int_0^t v_0 e^{-Kt/m} dt$$

$$x = (m/K)v_0(1 - e^{-Kt/m})$$

$$x_{\max} = mv_0/K$$

$$6. (1) dM = r\mu g dm \quad dm = \frac{m}{\pi R^2} dS = \frac{m}{\pi R^2} 2\pi r dr$$

$$M = \int dM = \int_0^R \frac{2\mu mg}{R^2} r^2 dr = \frac{2}{3} \mu mg R$$

$$(2) M \cdot t = J\omega - J\omega_0 \quad Mt = \frac{2}{3} \mu mg R t = \frac{1}{2} m R^2 \omega \quad t = \frac{3R\omega}{4\mu g}$$

(3) 转盘所受驱动力大小等于摩擦力大小, 驱动力所做的功为

$$W_{\text{驱}} = M\theta = \frac{2}{3} \mu mg R \omega t = \frac{1}{2} m \omega^2 R^2$$

摩擦力对唱片所做的功为 $W_1 = \frac{1}{2} J \omega^2 = \frac{1}{4} m \omega^2 R^2$

摩擦力对转盘所做的功为 $W_2 = -M\theta = \frac{2}{3} \mu mg R \omega t = -\frac{1}{2} m \omega^2 R^2$

一对摩擦力所做的总功为 $W = W_1 + W_2 = -\frac{1}{4} m \omega^2 R^2$